

Turn Battle MARL Study Report

OpenSpiel Experiment Pipeline

June 10, 2026

1 Overview

This report summarizes training and bracket-tournament results on the `turn.battle` OpenSpiel environment (4-player team combat).

2 Training curves

Figure 1 shows evaluation win rate against a random baseline during training.

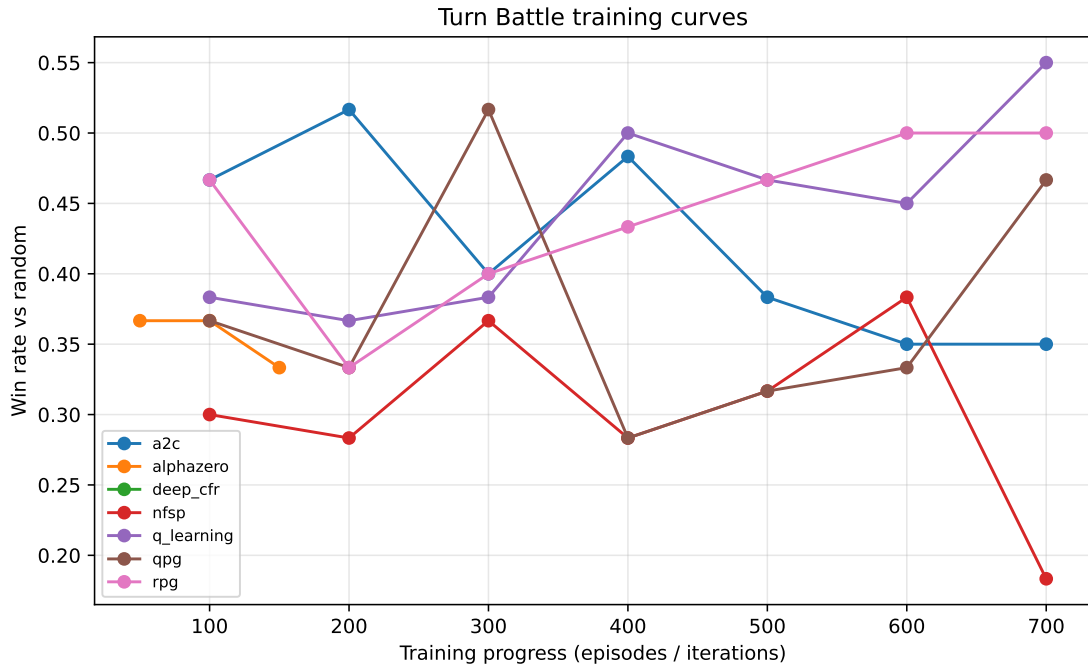


Figure 1: Training progress per algorithm.

3 Seeding

Algorithms were ranked by win rate vs. random before the bracket. MCTS, AlphaZero, and Deep CFR were seeded but excluded from the elimination bracket because they require turn-based evaluation.

Algorithm	Win rate vs random
alphazero	0.617
qpg	0.483
deep _{cfr}	0.250
nfsp	0.233

Table 1: Seeding scores for all algorithms.

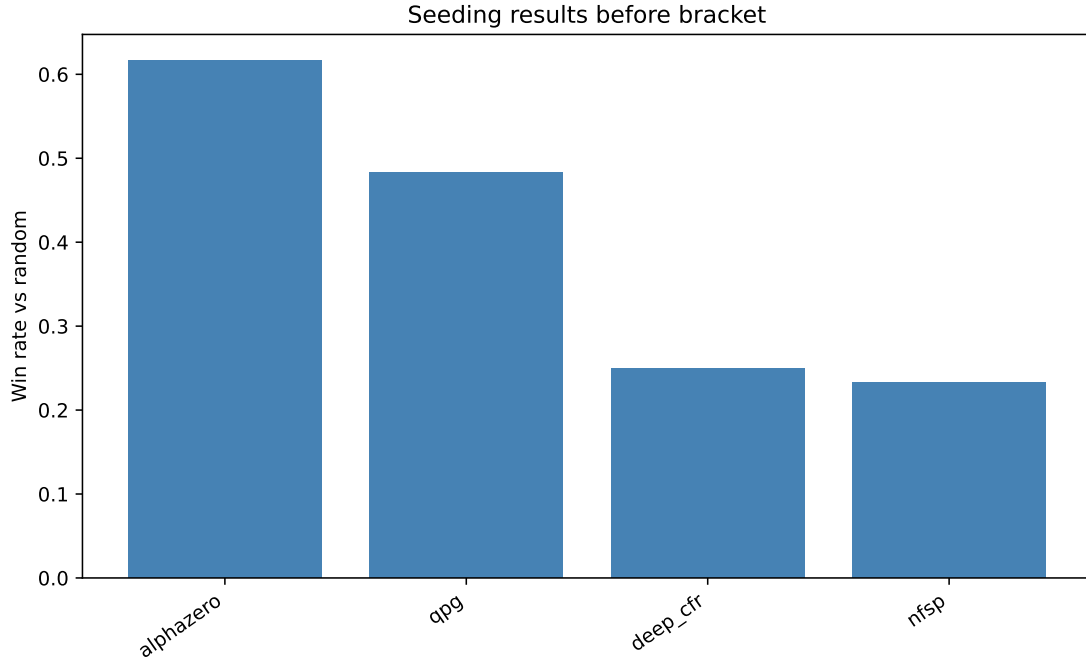


Figure 2: Seeding scores.

4 Bracket tournament

The champion was **alphazero**.

5 Conclusion

The bracket winner was **alphazero**, selected after full training and seeded single-elimination matchups.

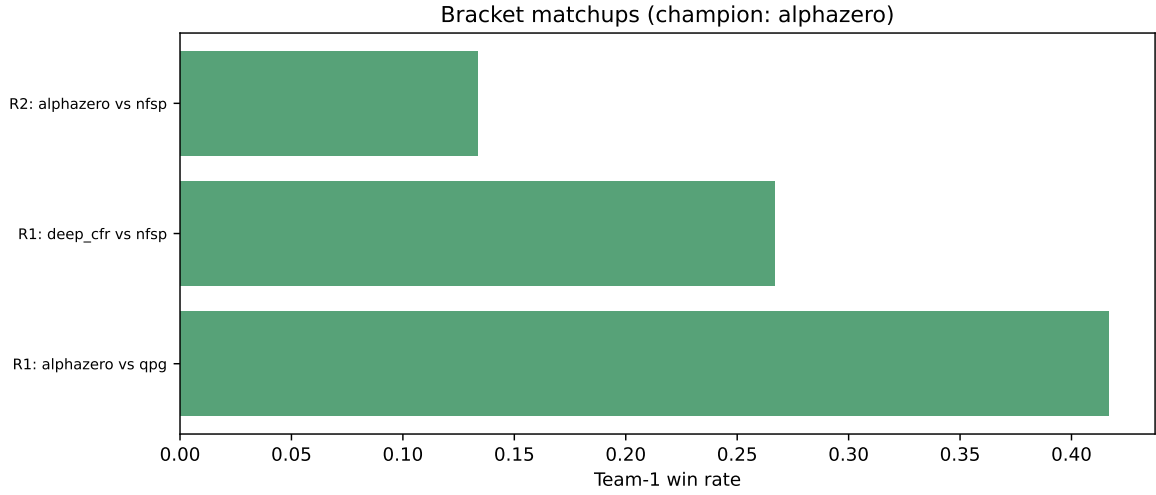


Figure 3: Head-to-head bracket results.

Round	Team 1	Team 2	Winner
R1	alphazero	qpg	alphazero
R1	deep _{cfr}	nfsp	nfsp
R2	alphazero	nfsp	alphazero

Table 2: Bracket match log.